

Sri Sai Raj

Aspiring Quantum Software & AI Engineer

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SUMMARY

Dual-degree student in Data Science and AI, building a foundation at the intersection of **quantum computing, deep learning, and NLP**. I learn by building: I'm developing a physics-informed neural network research paper and leading a team on a RAG-based document chatbot. I'm looking for a first internship where I can contribute clean, well-reasoned code, ask the right questions, and grow fast alongside a strong team.

PROJECTS & RESEARCH

Physics-Informed Neural Networks (PINNs) Research Paper

RaSoR Club, IIT Madras BS | Active ideation / early-stage development

Tech: Python, Deep Learning, Neural Networks, Differential Equations, Jupyter Notebooks

- Developing a research paper on **PINNs**, models that embed physical laws such as differential equations directly into the neural network loss function, so predictions respect the underlying physics.
- Studying how PDE residuals are turned into loss terms and combined with data-driven training, building on **linear algebra and calculus** foundations.
- Practicing the full research workflow: reading primary literature, breaking papers down into first principles, and shaping an original research direction within a collaborative subgroup.

NotebookLM-Style Document Chatbot (RAG)

B.Tech Team Project, SVCT | Team Lead (4 members) | Early-stage development

Tech: Python, NLP, LLMs, Retrieval-Augmented Generation (RAG), Prompt Engineering

- Building a **personalized, local chatbot** that parses uploaded PDFs and answers questions grounded in their content, in the style of Google's NotebookLM.
- Own the **backend and RAG pipeline** (document parsing, retrieval, LLM prompting) and the research design, including a focus on evaluating answer faithfulness to source documents.
- Leading a four-person team on architecture and planned features (PPT generation, short-note PDFs, question paper generation), sharpening my **system design, prompt engineering, and collaboration** skills.

TECHNICAL SKILLS

Languages	Python, SQL (MySQL)
AI / Machine Learning	Deep Learning, Neural Networks, Physics-Informed ML, NLP, LLM Prompt Engineering, RAG
Quantum Computing	Qiskit (exploring), Superposition & Entanglement, Quantum Mechanics fundamentals
Mathematics	Linear Algebra, Differential Equations (applied in PINNs)
Developer Tools	Git, GitHub, Jupyter Notebooks

EDUCATION

B.S. in Data Science and Applications Indian Institute of Technology, Madras	Expected 2029
B.Tech. in Artificial Intelligence & Data Science Sri Venkateswaraa College of Technology (SVCT), Chennai	Expected 2029

RELEVANT COURSEWORK & SELF-DIRECTED LEARNING

- Academic:** IITM BS Data Science foundational program (Python, data science fundamentals); B.Tech. AI & Data Science curriculum.
- Self-study (in progress):** Quantum computing fundamentals with hands-on IBM Qiskit; deep learning and neural network foundations; NLP and LLM application development.
- Approach:** Break down research papers from first principles, build concept-first, and turn what I learn into working projects on GitHub.

AREAS OF INTEREST

Quantum Software • Quantum Machine Learning (QML) • Data Science